

Daniel RACOCEANU
Professor

Sorbonne University
Faculty of Science and Engineering
Engineering and Computer Science Department
4 place Jussieu, 75005 Paris, France
e-mail: daniel.racoceanu@sorbonne-universite.fr
https://www.sorbonne-universite.fr/en

Paris Brain Institute / Institut du Cerveau • ICM
Center for Artificial Intelligence and Data Science • ARAMIS Lab
187 rue du Chevaleret, 75013 Paris, France
Tel.: +33 1 57 27 44 57 • e-mail: daniel.racoceanu@icm-institute.org
https://institutducerveau-icm.org/en/

BIOGRAPHY

Daniel Racoceanu is Professor in Biomedical Imaging, Pattern Recognition, Machine and Deep Learning, and Information/Data Analytics at Sorbonne University, and Principal Investigator at the Paris Brain Institute, within the Center for Artificial Intelligence and Data Science. His research focuses on computational pathology, spatial transcriptomics, biomedical image analysis, multimodal representation learning, and explainable artificial intelligence, with a particular emphasis on clinically meaningful and trustworthy AI systems.

He received his Ph.D. in 1997 and his Habilitation (HDR) in 2006 from the University Marie and Louis Pasteur, Besançon, France. In 1999, he joined the University Marie and Louis Pasteur as Associate Professor and Research Fellow at the FEMTO-ST Institute (CNRS UMR 6174).

Between 2005 and 2014, he contributed to the establishment and development of the International Joint Research Unit IPAL (UMI CNRS 2955) in Singapore, serving as its Director from 2008 to 2014. This international research structure was created through a collaboration between Sorbonne University, the French National Centre for Scientific Research (CNRS), the National University of Singapore (NUS), and the Agency for Science, Technology and Research (A*STAR). During this period, from 2009 to 2015, he was also Professor at the School of Computing at the National University of Singapore.

From 2014 to 2016, he served on the Executive Board of the University Institute of Health Engineering at Sorbonne University and was co-Director of the Cancer Theranostics research team at the Bioimaging Lab (LIB – CNRS UMR 7371, Inserm U1146).

From 2016 to 2018, he was Professor in Biomedical Imaging and Data Analytics at the Pontifical Catholic University of Peru.

Since 2020, he has been a member of the Advisory Board of the European Society for Digital and Integrative Pathology (ESDIP). He previously served as Vice-President from 2016 to 2018 and President from 2018 to 2020.

From 2018 to 2022, he served on the Board of Directors of the Medical Image Computing and Computer-Assisted Intervention Society (MICCAI). He was General Chair of MICCAI 2020 and contributed to the organization of MICCAI 2021 and MICCAI 2022.

His current work explores the integration of digital histopathology, spatial omics, and foundation AI models for next-generation précision medicine.

PROFESSIONAL & RESEARCH EXPERIENCE

Since May 2019 – **Sorbonne University, Paris Brain Institute (ICM)**, Paris, France

- **Principal Investigator, Paris Brain Institute (Institut du Cerveau – ICM)** - Inserm, CNRS, AP-HP
- **Center for Artificial Intelligence and Data Science**, Aramis Lab
- **Full Professor** @ Faculty of Science & Engineering, Engineering & Computer Science Department
    - Courses & Labs:
        - Artificial Intelligence, Machine / Deep Learning (master – Eng.) – coordinator
        - Medical and Biomedical Image Analysis (master level – Eng.) – coordinator
        - Biomedical Image Analysis (master level – Eng.) – coordinator
        - Computer Graphics (master level – Computer Sciences) – coordinator
        - Object Oriented Programming (Master level – Eng.)
        - Programming (undergraduate level – all options)
- Since 2020 – **Advisory Board** member - **ESDIP**[1] (European Society of Digital Integrative Pathology)
- 2023-2026 – Member of the **Inria Evaluation Committee** - **CE Inria** - **National Institute for Research in Digital Science and Technology**.
- 2027 – Chair of the ANR CE45 scientific panel - Interfaces: mathematics, digital sciences – biology, health (217 proposals).
- 2020-2026 – PI of the Inria team-project Aramis.
- 2020 – general Chair **MICCAI** 2020[2] (Medical Image Computing and Computer Assisted Intervention)
- 2018-2022 – Member of the **Board of Directors** of the international **MICCAI Society**
- 2018-2020 – **President of ESDIP** – the European Society of Digital Integrative Pathology

2016 – 2018: **Pontifical Catholic University of Peru** (PUCP), San Miguel, Lima, Peru

- **Exceptional Professor @** Faculty of Science and Engineering, Department of Engineering;
- Master and PhD students' supervision @ the Medical Imaging Lab;
- European & international initiatives around Computational and Integrative Pathology;
- Succeeding to the **MICCAI 2020** bid - Medical Image Computing & Computer Assisted Intervention (MICCAI 2020 conference was hosted for the first time by Latin America);
- 2018 – Member of **MICCAI Society Board of Directors** (Medical Image Computing and Computer Assisted Intervention)

2011- 2016: **University Pierre and Marie Curie**, Paris, France

- **Full Professor** @ Engineering Department.
- 2014-2016 - **Research Director** - cancer diagnosis and therapies team, **Biomedical Image Lab** (LIB[3] - CNRS UMR 7371, Inserm U1146);
- 2008-2014 – **Director** of the **CNRS** research lab **IPAL** (UMI CNRS 2955) **Singapore**
- **Executive Board**, **University Institute of Health Engineering (IUIS) of Sorbonne University**
- Co-Director of a new Bachelor of Sciences minor: "Innovation in Public Health"
- 2016 – co-founder of the **ESDIP** (European Society of Digital Integrative Pathology)
- 2016-2018 – Vice-president of the **ESDIP** (Eur. Society of Digital Integrative Pathology).

2013 – 2015**: National University of Singapore** (**NUS**), Singapore

- **Full Professor** (adjunct) @ School of Computing, Computer Science Dpt.
- 2008-2014 – **Director** of the **CNRS** research lab **IPAL - UMI CNRS 2955 Singapore**
- NUS SoC PhD co-supervision; NUS SoC PhD reviews and jury member.

2011 – 2014**: French National Center for Scientific Research (CNRS)**

- **CNRS Research Director in Biomedical Engineering and Imaging**
- Second mandate as **IPAL UMI CNRS 2955 Director**.

2008 – 2014: **Director**, **International Joint Research Lab IPAL UMI CNRS**, Singapore

- Restructuration around two impactful areas: biomedical image understanding and ambient assisted living; successful CNRS et al. evaluations in 2009 and 2014; growth of the lab by bringing in the University Pierre and Marie Curie and Mines-Télécom Institute.

2009 – 2013**: National University of Singapore (NUS)**, Singapore

- **Associate Professor** (adjunct) @ **School of Computing**, Computer Science Dpt.

---

[1] *ESDIP (European Society of Digital Integrative Pathology) : https://www.esdipath.org/*
[2] *MICCAI 2020: https://www.miccai2020.org/en/ORGANIZING-COMMITTEE.html*
[3] *Lab. Imagerie Biomédicale, LIB - UMR CNRS UMR7371, INSERM U1146: https://www.lib.upmc.fr/en/presentation/*

- NUS SoC PhD students' co-supervision, PhD reviews and PhD jurys
- Co-PI of the A*STAR SERC project "MMedWeb"
- Collaborator in a 2011 Singapore - MIT Alliance (SMA3)

2005 – 2011: **French National Center for Scientific Research (CNRS)**

- **CNRS Research Fellow (Chargé de Recherche CNRS),** creation and management of the Medical Image Indexing team at the International Research Unit PAL (UMI CNRS).
- Research focusing on Medical Image Analysis, Content Based Image Retrieval, Medical Ontologies, Medical Multimedia Fusion, Cognitive Vision.
- Participation @ CLEF international benchmark, Medical Image Indexing-Retrieval track.

1999 – 2008: **Marie and Louis Pasteur University** (former Franche-Comté Univ), Besançon, France

- **Associate Professor,** Faculty of Sciences & Technologies, Sciences & Technologies Dpt.
- Courses at Sciences and Techniques Dpt. in artificial intelligence, computer vision, diagnosis, prognosis, control sciences, fault detection, real-time, non-linear, and discrete event systems.
- Involved in the creation of the new LMD (Licence-Master-Doctorat) program.
- Research Fellow at **FEMTO-ST Institute**[4] (UMR CNRS 6174), Besançon, France, in AI applied to dynamic monitoring, diagnosis, and prognosis, in the e-maintenance framework.

1994 – 1996: **Marie & Louis Pasteur University**, Institute of Technology nord Franche-Comté, France

- **Lecturer** (ATER - Attaché Temporaire d'Enseignement et de Recherche)
- Dpt. Production System Management, University Institute of Technology of Belfort, University of Franche-Comté, Belfort, France
- Logistic management, supply chain management courses and laboratory
- Flexible production system labs (RFID, PLC programming, production flow management); I created a set of hands-on laboratories on a state-of-the-art flexible industrial production system.

1992: **Research Institute for Welding and Material Testing**[5] (ISIM), Timisoara, Romania

- **Research Fellow** in welding robotics.

PROFESSIONAL ORGANIZATIONS

Medical Image Computing and Computer Assisted Intervention (MICCAI Society)

- 2018-2022: member of the Board of Directors and General Chair of MICCAI 20204

European Society of Digital and Integrative Pathology (ESDIP3)

- Since 2020 - ESDIP Advisory Board
- 2018 to 2020 - President & Member of the Executive Committee
- 2016 to 2018 - Vice-President & Member of the Executive Committee

MOBILITY ACROSS THEMATIC AREAS

Since 2016	Integrative Digital Pathology & XAI - Prof. @ Sorbonne University and Prof. @ PUCP
2012-2016	Computational Pathology – Prof. @ UPMC & Prof. @ NUS & DR @ CNRS
2008-2012	BioMedical Imaging, Cognitive BioMedical Image, Stochastic Prior shape and Simplicial models – Prof. @ UPMC, Prof. @ NUS, DR & CR @ CNRS
2005-2008	Content-Based Medical Image Indexing and Retrieval reinforced by Semantic Approaches - A/Prof. @ UFC, A/Prof. @ NUS, CR @ CNRS
1999-2005	Dynamic Pattern Recognition for Diagnosis/Prognosis using dynamic recurrent neural networks - A/Prof. @ Marie and Louis Pasteur University (UMLP).
1992-1997	Stochastic Models Simplification and Control using Principal Component Analysis, PhD candidate @ UFC, Lecturer @ Institute of Technology, UMLP, Belfort, M.Sc. @ UTBM.
1987-1992	Mechatronics - M.Eng. @ "Politehnica" University of Timisoara (UPT).

GEOGRAPHIC MOBILITY

Since 2019	Paris, France - Prof. @ Sorbonne University, PI @ Paris Brain Institute (ICM)
2016-2018	San Miguel, Lima, Peru - Prof. @ Pontifical Catholic University of Peru (PUCP)
2014-2016	Paris, France - Prof. @ University Pierre and Marie Curie (UPMC Univ Paris 6)
2005-2014	Singapore - CNRS Senior Research Fellow & UMI CNRS Lab Director
1999-2005	Besançon, France - A/Prof. @ Marie and Louis Pasteur University (U. Franche-Comté)

[4] *FEMTO-ST Institute (UMR CNRS 6174), Besançon, France: https://www.femto-st.fr/en*
[5] *National R&D Institute for Welding and Material Testing (ISIM), Timisoara, Romania: https://www.isim.ro/en/*

1993-1997 Belfort, France – PhD candidate @ University of Franche-Comté & Lecturer (ATER) @ Belfort-Montbéliard University Institute of Technology (IUT)
1992-1993 Belfort, France - Master of Sciences @ Univ. of Technology Belfort-Montbéliard (UTBM)
1987-1992 Timisoara, Romania - Master of Engineering @ Politehnica Univ. Timisoara (UPT)

EDUCATION

2006 **HDR (Accreditation to Supervise Research / Habilitation à Diriger des Recherches)**
- Control and Computer Science

University of Franche-Comté, Besançon, France
- Dynamic Monitoring using Artificial Intelligence Techniques

- *Keywords:* Dynamic Monitoring, Artificial Intelligence, Dynamic Neural Networks, Neuro-Fuzzy Systems, Fuzzy Petri Nets, Diagnosis, Prognosis.

1997 **Ph.D. Control and Computer Sciences, University of Franche-Comté, Besançon, France**
- Stochastic model reduction. Quasi-optimal management solution of an EDF (Electricity of France) Hydropower System (a system with 7 centrals).
- *Keywords:* Stochastic Modelling, Markov Chains, Control, Reduction Methods, Singular Perturbations - Principal Component Analysis.

(Highest distinction - jury's congratulations – très honorable avec félicitations du jury).

1993 **M.Sc. (Master of Science) Control Sciences**
University of Technology of Belfort-Montbéliard, France
(Major & with Distinction – Très Bien).

1992 **Dipl. Ing. (Master of Engineering) - Mechanical Manufacturing Engineering Technology**
Politehnica University of Timisoara, Romania
(Highest Distinction in Romanian academic system – 10/10).

CURRENT SCIENTIFIC ACTIVITY

General interest: Responsible Artificial Intelligence, High-content Biomedical Image analysis
Responsible Artificial Intelligence, Explainable AI, Computer science for biomedical data, information and knowledge analysis / management. Integrative biomedical image analysis.

Specific area of expertise: Explainable Artificial Intelligence / Deep Learning, Integrative approaches for biomedical data analysis, Semantic-driven high-content image exploration
Explainable Artificial Intelligence / Deep Learning, Integrating heterogeneous models for omics and high-content imaging data understanding, with initial (not exclusive) focus on digital pathology. Semantic-driven, traceable deep-learning and scalable image analysis.

Keywords: High-content biomedical image analysis, integrative digital pathology, semantic-driven high-content biomedical image analysis, explainable and responsible deep learning / artificial intelligence, stochastic modelling and prior shapes, mathematical morphology on sparse-sets, content-based image retrieval.

SCIENTIFIC PUBLICATIONS

Exhaustive publication list: https://daniraco.github.io/publications.html

Google Scholar: https://scholar.google.com/citations?user=2eBRLj0AAAAJ&hl=fr
Research Gate: https://www.researchgate.net/profile/Daniel_Racoceanu
ORCHID: http://orcid.org/0000-0002-9416-1803
Web of Science Researcher-ID: F-4576-2011
DBLP profile: https://dblp.org/pid/63/4416.html
Semantic Scholar ID: https://www.semanticscholar.org/author/Daniel-Racoceanu/1742773

SCIENTIFIC SUPERVISION

- **Co-guarantor Dr. Habil – habilitation to manage scientific research (HDR)**
  - Dr Nicolas LOMENIE, A/Prof, Univ Paris Descartes, IPAL Res. Fellow Jan.2009-Sep.2012
- **Post-docs supervised**
  - Dr Anuradha KAR (ICM, 2021- 2023) - Alzheimer's disease patients' stratification
  - Dr Janan ARSLAN (ICM, 2021-2023)- model. metabolic plasticity heterogeneity in melanoma
  - Dr Pawan KUMAR (2024, co-supervised with O. Radulescu) - data-driven mechanistic modelling of oxygen distribution and hypoxia in the tumour microenvironment

- Dr Antoine VEILLARD (UPMC, 2013-2016) - collaborative digital pathology
- Dr Ludovic ROUX (UJF, 2008-2014) - semantic cognitive approaches
- Dr HUANG Chao-Hui (NUS and A*STAR, 2008-2014) - cell tracking and WSI analysis.
- Dr Maria KULIKOVA (CNRS/UPMC 2010-2013) - marked point process, nuclei detection.
- Dr Caroline LACOSTE (CNRS, 2005-2006) – marked point process angiography analysis

- **On-going and graduated PhDs:**

- *On-going PhD supervision:*
  - M. Mehdi HAMADACHE (start Oct. 2025, Sorbonne Univ., 100%) – spatial transcriptomics using physically inspired AI.
  - Ms. Swann RUYTER (start Oct. 2024, Sorbonne Univ., 100%) – virtual staining, virtual proteomics and virtual spatial transcriptomics using generative IA.
  - Ms. Esther KOZLOWSKI (start Oct. 2023, Sorbonne Université, 50%) – analysis of propagation pathways in Parkinson's disease using explainable AI.
- *Graduated PhDs supervised / first position after PhD:* http://www.theses.fr/fr/?q=racoceanu
  - M. Ilias SARBOUT (Oct. 2026, Sorbonne Université, Paris) – collaboration with Adolphe de Rothschild Foundation Hospital - Measuring, Preventing, and Compensating for Vision Loss: Machine Learning and Digital Twins under Data Constraints.
  - Ms. Ayse GUNGOR (Sept. 2026, Sorbonne Université, Paris) – collaboration with Adolphe de Rothschild Foundation Hospital - Artificial Intelligence in Neuro-Ophthalmology: From Automated Diagnosis to Large-Scale Retinal Dataset Curation.
  - Ms. Laura MARIN (2023, PUCP, Lima, Pérou) – Digital Integrative Pathology for personalized medicine, R&D engineer - AREAS Grenoble, France.
  - M. Mehdi OUNISSI, (Oct. 2024, Sorbonne Univ., Paris) - Explainable Artificial Intelligence, post-doc researcher at the Adolphe de Rothschild Foundation Hospital, Paris, France.
  - M. Gabriel JIMENEZ, (Sept. 2024, Sorbonne Univ., Paris) - Interpretable Deep Learning in Computational Histopathology for Alzheimer Disease Patients' Stratification Refinement, Research engineer at Paris Brain Institute, Paris, France.
  - Ms. Oumeima LAIFA, (Sept. 2019, Sorbonne Univ., Paris) - A Joint Discriminative-Generative Approach for Tumor Angiogenesis Assessment Comp. Pathology, R.Eng.Owkin.
  - M. Lamine TRAORE, (Dec. 2017, UPMC, Paris) - Semantic Modelling of a Histopathology Image Exploration & Analysis Tool, CEO smart'GRAD – UPMC spin-off.
  - M. Bassem BEN CHEIKH (Sept. 26, 2017, UPMC, Paris) - Graph-based Mathematical Morphology for Characterizing the Spatial Organization of Histological Structures in High-Content Images: Appl. Tumor Microenvironment in Breast Cancer.
  - M. Olivier MORERE (June 2016, UPMC, Paris) - Deep Learning Compact and Invariant Image Descriptors for Instance Retrieval.
  - Ms. Sreetama BASU (March 2015, NUS, Singapore) - Digital Reconstruction of Neuronal Structures from 3D Microscopy data / Research Fellow @ École Normale Supérieure (Ulm) Paris, Institute of Biology, Bioinformatics & Comput. Biology group;
  - M. Antoine FAGETTE (June 2014, UPMC, Paris) - Dense Crowd Analysis / Research Engineer THALES Singapore;
  - M. Stéphane RIGAUD (March 2014, UPMC, Paris) - Analysis-Synthesis Approach for Neurosphere Modelisation Under Phase-Contrast Microscopy / 3 years Research Fellow @ Institut Curie, Paris;
  - M. Humayun IRSHAD (Jan. 2014, UJF, Grenoble) - Automated Mitosis Detection in Color and Multispectral High-Content Images in Histopathology: Appl. to Breast Cancer Grading in Digital Pathology / Res. Fellow (3 years contract) @ Harvard Medical School, Boston, US
  - M. Antoine VEILLARD (Dec. 2012, NUS, Singapore) - Kernel Methods for the Incorporation of Prior-Knowledge into Support Vector Machines. Application to whole slide image analysis for breast cancer grading / Research Fellow (3 years) @ Univ. Pierre & Marie Curie, Paris;
  - Ms. Roxana TEODORESCU (Apr. 2011, Univ. Besançon) - Parkinson's Disease Prognosis using Diffusion Tensor Imaging Features Fusion / Research Fellow (3 years) @ Mount Sinai Hospital, New York, USA.
  - Ms. Adina TUTAC (Oct. 2010, Univ. Besançon) - Formal Representation and Reasoning for Microscopic Medical Image-Based Prognosis. Application to Breast Cancer Grading / R&D Manager @ Sionic SRL, Timisoara, Romania.
  - M. Nicolas PALLUAT (Jan. 2006, Univ. Besançon) - Dynamic monitoring using temporal neuro-fuzzy syst. / Res. Fellow (3 years contract) @ Univ. Federal Santa Catarina, Brazil;

- Ms. Eugenia MICA (Sept. 2004, Univ. Besançon) - Discrete events systems monitoring using fuzzy Petri nets / A/Prof @ Univ. Valahia, Târgoviste, Romania.
- M. Ryad ZEMOURI (Nov. 2003, Univ. Besançon) - Monitoring using dynamic neural networks A/Prof. CNAM, Paris, France.

SUCCESSFUL FUNDRAISING

Major competitive projects:

- **Interdisciplinary approaches in oncogenic processes and therapeutic perspectives: Contributions of mathematics and informatics to oncology – MIC 2025 - Synergizing Mechanistic and AI Approaches for Modeling the Impact of Microenvironment Heterogeneity on Immunotherapy efficacy (SIMAI)**
  - Partners: Paris Brain Institute (CNRS UMR7225 – Inserm U1127), the University of Montpellier (LPHI - CNRS UMR5235), and the Montpellier Cancer Research Institute - IRCM (Inserm U1194).
  - Role: PI. Personal contribution: multimodal biomedical imaging analysis using frugal AI
  - Duration: 4 years (Oct. 2025 – Sept. 2029). Budget: 528 400 €.

- **Big Brain Theory (BBT3), Paris Brain Institute - STRATIFIAD: Refining Alzheimer Disease Patients' stratification using effective, traceable and explicable artificial intelligence approaches in computational histopathology.**
  - Leader of the project. Personal contribution: WSI analysis and XAI / DL approaches
  - Partners: INRIA team Aramis lab, Alzheimer team, and Data analysis center, Paris Brain Institute
  - Duration: 3,5 years including a one-year extension (July 2021 - December 2024). Budget 200 000 €.

- **AVIESAN (French Alliance for Life and Health Sciences) - ITMO Cancer - Mathematical Approaches to Modelling Metabolic Plasticity and Heterogeneity in Melanoma (MALMO)**
  - Partners: Paris Brain Institute (CNRS UMR7225, Inserm U1127), University of Montpellier (LPHI CNRS UMR5235, LIRMM (CNRS UMR 5506) and Montpellier Cancer Research Institute - IRCM (Inserm U1194)
  - Role: PI. Personal contrib.: traceable DL for 3D histological reconstruction and tumor characterization
  - Duration: 4 years including a one-year extension (Nov. 2020 – Oct. 2024). Budget: 400 000 €

- **Medical Research Foundation (FRM, France)**; Bioengineering for health
  - **Multiparametric ultrasonic classification to evaluate tumor progression**
  - Partners: Sorbonne Université, Hôpital Pitié-Salpêtrière, Univ. Illinois Urbana-Champaign
  - Duration: 3 years (2014-2017). Role: PI. Personal contribution: WSI analysis using DL

- **NIH (Nat. Institute Health, US) - Advanced Ultrasonic Evaluation of Sentinel Lymph Nodes**
  - Consortium: Riverside Research, New York NY USA, Hawaii University, Honolulu, USA, Kuakini Medical Center, Honolulu, USA, Duration 5 years (2011-2016);
  - Role: collaborator. Personal contribution: WSI analysis for the classification of metastatic regions in lymph nodes, in correlation with ultrasound modalities.

- **IAMS: Integrated autonomous microscopy syst. for imaging anatomies complex 3D cell**
  - Granted by **A*STAR/JCO** Joint Council Office SERC-BMRC/A*STAR Singapore;
  - Involving IPAL and four A*STAR institutes (the Institute for Infocomm Research I2R/SERC, the Bioinformatic Institute BII/BMRC, the Institute of Molecular and Cell Biology IMCB/BMRC and the Institute of Medical Biology IMB/BMRC);
  - Duration: 3 years (2013 – 2016). Role: PI. Personal contribution: microscopic images analysis.

- **MICO: COgnitive MIcroscope: Cognition-driven visual explorer in histopathology**
  - Granted by the Techn. for Health program (TecSan - Technologies pour la Santé) of the **French National Research Agency** (ANR - Agence Nationale de la Recherche);
  - Partners: LIP6 -UPMC, TRIBVN, THALES-TCF, Pitié-Salpêtrière Hospital, AGFA Healthcare;
  - Ontology- and semantics-driven exploration of whole-slide images; proof of concept of semantics-driven breast cancer grading.
  - Duration: 3,5 years (01/2011-07/2014). Grant: 1.100.000 €.
  - Personal role: **Project Director & Principal Investigator**

- **Intel. Vision Quantitative Microscopy Neural Stem Cells Progenitor Growth Differentiation**
  - Project funded by **A*STAR/JCO** (Joint Council Office SERC-BMRC/A*STAR) Singapore Involving three A*STAR institutes (Institute for Infocomm Research I2R/SERC, Bioinformatics Inst. BII/BMRC and the Inst. of Medical Biology IMB/BMRC);
  - Duration: 3 years (2009 – 2012). Grant: 375.000 €. Personal role: PI.

- **C0604 - EURO-TELEPATH - "Telepathology Network in Europe"**
  - **COST (European Cooperation in Science and Technology)** Action.
  - Partners: Spain, France, Germany, Grece, Italy, Switzerland, Croatia, Finland, Lithuania, Nederland,

Norway, Poland, Portugal, United Kingdom;
  - Duration: 4 years (2007-2011). Personal role: expert in semantic imaging

- **MMedWeb (Multimedia Medical Conceptual Web for Intelligent Information Access)**
  - **A*STAR – SERC** grant - Science & Engineering Research Council, Agency for Science, Technology and Research – Singapore.
  - Partners: CNRS, NUS, I2R/A*STAR, Nat. Univ. Hospital (NUH), National Healthcare Group, Singapore General Hospital (SGH);
  - Duration: 3 years (2007-2010). Grant: 250.000 €. Personal role: co-PI.

- **ONCO-MEDIA (Ontology and COntext related MEdical image Distributed Intel. Access)**
  - 6th **ICT-ASIA programme**
  - 11 International partners: Singapore (IPAL), France (CREATIS, LIRIS, LIP6, I3S), Switzerland (UNIGE), Taiwan (NTU), Philippine (ATENEO), Japan (CIGG);
  - Duration of the project: 2 years (2006-2007; 2009-2010); Budget: 100.000 €;
  - Personal role: **Project Director & Principal Investigator**

AWARDS:

- 1st Prize – Poster contest – European Neuro-Ophtalmology Society EUNOS 2024, Rotterdam, Netherlands - Gungor, A., et al., Early Detection of Central Retinal Artery Occlusion within 4.5 Hours of Visual Loss: Deep Learning Method Applied on Fundus Photographs.
- Best Scientific Platform Presentation Prize – European Neuro-Ophthalmology Society EUNOS 2026, Milan, Italy - Gungor, A., et al.
- 1st Place Prize - 2019 IEEE Eng. in Medicine and Biology Prize: Irshad, H. et al. (2014). Methods for Nuclei Detection, Segmentation, and Classification in Digital Histopathology: A Review - Current Status & Future Potential. *IEEE Reviews on Biomedical Engineering.*
- Best Paper Award finalist - Zemouri, R., Racoceanu, D., Zerhouni, N. (2001). A Petri nets graphic method of reduction using birth-death processes, *IEEE International Conference on Robotics & Automation - IEEE ICRA'2001, pp. 46-51, Seoul, Korea.*

PATENTS:

Patent / Software (Title & Reference)	Details & Role
Virtual staining – paired images *Device and method for generating n virtual IHC stain images from one H&E image (paired img)* EP 24 305 221.4 (Europe, filed 9 Feb 2024) PCT/EP2025/053151 (World filed 6 Feb 2025) WO2025/168731 (World published 14 Aug 2025)	Field: Computational pathology, virtual immunohistochemistry (IHC) Applicants: ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité Role: Co-inventor & institutional co-owner
Virtual staining – unpaired images *Device and method for generating n virtual IHC stain images from one H&E image (unpaired img)* EP 24 305 224.8 (Europe, filed 9 Feb 2024) PCT/EP2025/053153 (World filed 6 Feb 2025) WO2025/168729 (World published 14 Aug 2025)	Field: Computational pathology, virtual IHC Applicants: ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité Role: co-Inventor & institutional co-owner
3D Tracking in Microscopy *Method and system for tracking motion of microscopic objects within a 3-D volume* US 2014/0192178 A1, filed 13 Aug 2012, pub. 10 Jul 2014 (A*STAR Singapore)	Field: Biomedical imaging, 3-D cell tracking Role: co-Inventor (holder A*STAR)
Dynamic Real-Time Monitoring System INPI France 0858450, filed 10 Dec 2008 Reference: INPI ref. 8E50BTFR1-Em@systec (Em@systec France)	Field: Industrial vision, sensor systems Role: Co-inventor (holder Em@systec France - spin-off UFC)
Mitosis Detector CNRS DL 05963-01 (UMI 2955 IPAL, 2013)	Field: Computational pathology, cell event detection Role: co-Author (CNRS copyright)
PDFibAtl@s CNRS DL 04137-01 (UMI 2955, submitted 16 Dec 2010, approved 13 May 2011)	Field: Neuroimaging, MRI-DTI data fusion for Parkinson's disease Role: co-Author (CNRS copyright)

Patent / Software (Title & Reference)	Details & Role
HISTOGRAD CNRS DL 2944-01 (UMI CNRS 2955); APPIDDN.fr.001.190019.000.S.P.2010.000.20900 (14 May 2010)	Field: Digital pathology, breast cancer grading Role: co-Author (CNRS copyright)

CERTIFICATES:

- June 2014, Certif. in Advanced English (CAE), Cambridge English Language Assessment.